崑山科技大學日間部電機工程系四技綠能科技技優領航專班課程科目表
Kun Shan University 4-Year Green energy technology technical class College Curricula

課程名稱(講授時數—實習時數—學分數)

Course Name (Lecture Hours – Practice Hours – Credits)

114.04.17 系課程委員會通過
114.04.22 院課程委員會通過
114.07.22 校課程委員會通過
113 學年度入學適用

Apply to the freshmen enrolled in 113

第一學年 First Academic Year		第二學年 Second Academic Year		第三學年 Third Academic Year		第四學年 Fourth Academic Year	
第一學期 First Semester	第二學期 Second Semeste	第一學期 First Semester	第二學期 Second Semeste	第一學期 First Semester	第二學期 Second Semeste	第一學期 First Semester	第二學期 Second Semeste
共同必修科目(合計 29 學分) University-wide Required Courses (29 Credits)							
校訂共同科目(合計 2 學分) General Courses (2 Credits)							
	生活實踐教育(一) Living Practice (I) (0-1-1)	生活實踐教育(二) Living Practice (II) (0-1-1)	服務學習與公民教育 Public Service and Civil Education (0-1-0)				
核心通識科目(合計 17 學分) Common Core Courses (17 Credits)							
體育 (一) Physical Education (I) (2-0-2)	體育 (二) Physical Education (I) (2-0-2)			生涯發展與職業倫理 Career Development and Work Ethics (2-0-2)	外語選讀 Foreign Languages ** (2-0-2)		英文能力檢定* English Proficiency Examination* (1-0-1)
中文閱讀與表達(一) Chinese Reading and Expression(I) (2-0-2)	中文閱讀與表達(二) Chinese Reading and Expression(II) (2-0-2)						
英文聽力與閱讀(一) English Listening and Reading (I) (2-0-2)	英文聽力與閱讀(二) English Listening and Reading (II) (2-0-2)						
分類通識科目(合計 10 學分) Liberal Education Courses (10 Credits)							
分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)			

註：Note:

*「英文能力檢定」依「崑山科技大學學生英文基本能力檢定實施辦法」辦理。

'English Proficiency Examination' is implemented according to 'Kun Shan University Guidelines for English Proficiency Requirement.'

**「外語選讀」包括「第二外語」及「職場英文」為選項式必修科目，課程擇一門修習即可。課程開設於三年級，然「第二外語」課程為二年級優先選課。

'Foreign Languages' comprises 'Foreign Languages (other than English)' and 'Workplace English', from which students choose one course. While the courses are offered in the third academic year, 'Foreign Languages (other than English)' is given preference to sophomores.

專業必修科目(合計 30 學分) Required subjects for majors (Total：30 Credits)

工程技術概論 Introduction to Engineering Technology (2-0-2) 可程式控制器基礎及實習 Programmable controller base and internship (1-2-3)	程式設計 Programming (1-2-3)	電機機械實習 Electric Machinery Experiment (0-2-2)		實務專題(一) Practice Project (Ⅰ) (0-3-3) 電力電子學實習 Experiments of Power Electronics (0-2-2) 基礎量測及實習 The Fundamental of Measurement and Experiments (1-2-3)	實務專題(二) Practice Project (II) (0-3-3) 控制系統實習 ontrol systems Experiment (0-2-2)	學術倫理* Academic ethics (1-0-1)	就業與創業技能輔導 Employment and entrepreneurial skills counseling (3-0-3) 專業證照** Professional Licenses (1-0-1)	暑期校外實習*** Summer Internship (0-＊-2)
3-2-5	1-2-3	0-2-2	0-0-0	1-7-8	0-5-5	1-0-1	4-0-4	0-*-2

註*：「學術倫理」之實施與學分取得依課務組網頁之「學術倫理教育實施要點」辦理。The implementation of "Academic Ethics" and the acquisition of credits are handled in accordance with the "Implementation Points of Academic Ethics Education" on the website of the Academic Affairs Group.

註**：專業證照課程依電機系之「專業證照課程實施辦法」辦理。Professional certificate courses are handled in accordance with the "Implementation Measures for Professional Certificate Courses" of the Department of Electrical Engineering.

註***：暑期校外實習課程依電機系之「校外實習課程實施辦法」辦理。Summer off-campus internship courses are handled in accordance with the "Implementation Measures for Off-campus Internship Courses" of the Department of Electrical Engineering.

專業選修科目，依每學期實際情況開課 (選修學分 69 學分) Major elective subjects are taught according to the actual conditions of each semester (69 credits for electives)

普通物理與實驗(一) University Physics Experiment(I) (2-1-3) 微積分(一) Calculus(I) (2-0-2) 材料科學導論 Fundamentals of Materials Science and Engineering (3-0-3) 充電樁系統概論 Introduction to charging pile system （3-0-3） 車用晶片模組應用系統概論 Introduction to automotive chip module application system （3-0-3）	普通物理與實驗(二) University Physics Experiment(Ⅱ) (2-1-3) 微積分(二) Calculus(Ⅱ) (2-0-2) 電路學 Electric circuits (2-0-2) 邏輯設計 Logic Design (2-0-2) 邏輯設計實習 Logic Design Experiment (0-2-2) 可程式控制器應用及實習 Practice of Programmable Logic Controller and Internship (1-2-3) 綠色能源概論 Introduction to Green Energy (3-0-3) C++程式語言 C++ Programming Language (3-0-3)	微處理機基礎（3-0-3） Basis of Microprocessor 工程數學(一) Engineering mathematics (I) (3-0-3) 電子學與實習 Electronics and practice （1-2-3） 電路學(二) Electric circuits(II) (2-0-2) 電機機械 Electric Machinery Experiment (2-0-2) C#程式語言 C# Programming Language (3-0-3) 綠能應用概論 Introduction to Green Energy Applications (2-0-2) 科技英文 Technology English (2-0-2) 機器人工程師實務 Robot Engineer Practice (1-2-3)	電機機械專論 Advanced of Electric Machinery (3-0-3) 工程數學(二) Engineering mathematics (Ⅱ) (3-0-3) 進階電子學與實習 Advanced Electronics and practice (1-2-3) 太陽能電池 Solar Cell (3-0-3) 半導體綠能科技應用 Semiconductor green energy technology applications (3-0-3) 光電工程導論 Introduction to Optoelectronic Engineering (3-0-3) 半導體產業技術 Introduction to Semiconductor Technology (3-0-3)	電力系統 Power System (3-0-3) 可程式 VB 圖形監控 Visual VB Monitoring of Programmable Logic Controller (3-0-3) 電力電子學 Power Electronics (2-0-2) 可程式 LabView 圖形監控 LabView Visual Monitoring of Programming Logic Con (3-0-3) 真空技術 Vacuum technology (3-0-3) 高亮度 LED 技術 High brightness LED technology (3-0-3) 專利實務 Patent practice (3-0-3) 充電樁規劃與設計 Charging pile planning and design (3-0-3)	電力系統分析 Power System Analysis (3-0-3) 發變電工程 Engineering of Power Generation and Transformation (2-0-2) 照明技術 Micro LED (3-0-3) 控制系統 control systems (2-0-2) 太陽光電能發電系統 Photovoltaic Energy Generation System (3-0-3) 感測元件應用與實習 Sensing component application and internship (1-2-3) 風力發電技術 Wind Power Technology (3-0-3) 充電樁製造生產實務 Charging pile manufacturing and production practices (2-2-4)	電動機控制與實務 Motor Control and Practice (3-0-3) 電力品質分析 Electric Power Quality Analysis (3-0-3) 配電工程 Power Distribution System (2-0-2) 數位電子乙級技術士術科 Digital Electronics grade B technique master technique (1-2-3) 電力電子乙級技術士術科 **Skill subjects of** Level B Technician for Power Electronics (1-2-3) 薄膜技術實務 Thin Film Technology Practice (3-0-3) 學期校外實習(一) Semester Intership (I) (0-＊-9)	能源應用 Energy applications (3-0-3) 電力電子應用 Applications of Power Electronics (3-0-3) 電力系統模擬 Power System Simulation (3-0-3) 電力控制與能源管理 Power Control and Energy Management (3-0-3) 電力電子模擬 Power System Simulation (3-0-3) 真空半導體設備人機介面及實習 Practice on human interface of vacuum equipment (1-2-3) 學期校外實習(二) Semester Intership (Ⅱ) (0-＊-9) 充電樁測試檢驗實務 Charging pile testing and inspection practices (2-2-4)

技優生推薦選修之三創與跨領域課程 Recommended electives for gifted students: three innovative and cross-field courses
※三創選修課程(須修至少 18 學分) Three elective courses (required to take at least 18 credits)

創意程式設計-C-E creative programming -C-E (2-0-2) 創意媒體與生活 Creative media and life (1-0-1)	創意故事 creative story (2-0-2) 創意管理 creative management (3-0-3)	創意電子控制工具使用入門-E Getting Started with Creative Electronic Control Tools-E (2-0-2) 創意媒體英文表達-院核心 Creative Media English Expression-House Core (2-0-2) 創意文案撰寫 Creative Copywriting (1-2-3)	創意行銷 Creative Marketing (2-0-2) 創意開發 Creative Development (2-0-2) 電腦輔助創新設計 Computer-Aided Innovative Design (1-2-3)	綠色科技與創新-C Green Technology and Innovation-C (2-0-2) 創意思考與方法 Creative Thinking and Methods (2-0-2) 廣告創意與文案 Advertising Creativity and Copywriting (1-1-2)	創新事業導論-B Introduction to Innovation Business-B (2-0-2) 創新智慧生活科技 Innovative Smart Life Technology (2-1-3)	創意思考訓練 Creative Thinking Training (2-0-2) 創意與專利 Creativity and Patent (3-0-3) 創意與創業 Creativity and Entrepreneurship (3-0-3)	就業與創業技能輔導 Employment and Entrepreneurship Skills Counseling (3-0-3) 環保與創意 Environmental Protection and Creativity (3-0-3) 創業與經營實務 Entrepreneurship and Business Practice (2-1-3)
※跨領域選修課程(須修至少 18 學分) Interdisciplinary elective courses (at least 18 credits required)							
綠能科技與實作 Green Energy Technology and Implementation (2-1-3) 太陽能工程概論 Introduction to Solar Engineering (3-0-3) 燃料電池應技術 Fuel Cell Technology (3-0-3) 電子商務 E-Commerce (3-0-3)	工程材料學 Engineering Materials (3-0-3) 工程技術概論 Introduction to Engineering Technology (2-0-2) 遊戲企劃 Game Planning（3-0-3）	智慧車輛控制系統 Intelligent Vehicle Control System (3-0-3) 智慧製造技術 Smart Manufacturing Technology (3-0-3) 多媒體製作 Multimedia Production (3-0-3) 遊戲行為分析 Game Behavior Analysis (2-0-2)	物聯網與大數據於智慧製造應用 Internet of Things and Big Data in Smart Manufacturing Applications (3-0-3) 工業 4.0 特色技術 Industry 4.0 Featured Technologies (3-0-3) 智慧商務應用 Smart Business Applications (3-0-3) 智慧商圈管理 Smart Business District Management (3-0-3)	新能源與人文社會社會發展 New Energy and Humanistic and Social Development (2-0-2) 綠色能源概論 Introduction to Green Energy (3-0-3) 材料科學導論 Introduction to Materials Science (3-0-3) 綠建築與智慧建築 Green Building and Smart Building (3-0-3) 行動遊戲製作 Mobile Game Production (2-1-3)	儲能系統概論 Introduction to Energy Storage Systems (3-0-3) 綠色能源與產業商機 Green Energy and Industrial Business Opportunities (3-0-3) 企業節能減碳概論 Introduction to Enterprise Energy Saving and Carbon Reduction (3-0-3) 環境規劃與管理 Environmental Planning and Management (3-0-3) 能源應用 Energy Applications (3-0-3) 遊戲行銷營運實務 Game Marketing Operation Practice (0-2-2)	環工生質能源 Environmental Biomass Energy (3-0-3) 溫室氣體管理 Greenhouse gas Management (3-0-3) 光電材料 Optoelectronic Materials (3-0-3) 節能材料概論 Introduction to Energy-Saving Materials (3-0-3) 太陽光電能發電系統 Solar Photoelectric Power Generation System (3-0-3)	物聯網於綠能之應用 Application of Internet of Things in Green Energy (3-0-3) 智慧能源與物聯網 Smart Energy and Internet of Things (3-0-3) 物聯網與綠能概論 Introduction to Internet of Things and Green Energy (2-0-2) 物聯網與綠能應用概論 Introduction to Internet of Things and Green Energy Applications (3-0-3) 物聯網與綠能應用實務 Internet of Things and Green Energy Application Practice (3-0-3)	

註：

校外實習課程依電機系之「校外實習課程實施辦法」辦理

Off-campus internship courses are handled in accordance with the "Implementation Measures for Off-campus Internship Courses" of the Department of Electrical Engineering

最低畢業學分：128 學分 The minimum total credits for graduation is 128 credits

備註：

1. 本系畢業學分為 128 學分，必修 59 學分(包含共同科目、系專業必修科目)、選修 69 學分(系專業選修科目、三創選修課程、跨領域選修課程及學生自行選修經系/學位學程認可之專業課程)。The minimum graduation credits of this department are 128 credits: 59 credits of compulsory courses and 69 credits of elective courses
2. 本技優專班課程，規劃 18 學分以上的三創課程及跨領域等選修課程，協助技優生運用優秀的技術能力及創造力，呈現多元的創作能量，體現技職教育的特色。This special class for technical students plans more than 18 credits of three innovative courses and cross-field elective courses to help technical students use their excellent technical abilities and creativity to present diverse creative energy and reflect the characteristics of technical and vocational education.
3. 本系學生需至少修讀完成一跨院系學分學程，取得跨院系學分學程證書使得畢業。Students in this department need to complete at least one cross-department credit course and obtain a cross-department credit course certificate to graduate.。
4. 本系學生需至少修讀完成一數位科技微學程，取得數位科技微學程證書使得畢業。Students in this department need to complete at least one digital technology micro-course and obtain a digital technology micro-course certificate to graduate.
5. 選修跨院系學程之外系專業課程，得等同本系之專業選修學分。Elective professional courses outside the department's cross-department program will be equivalent to the professional elective credits of the department.

暑期校外實習之實施為 4 週 160 小時 2 學分於暑假期間實施，若超過 160 小時，第 161-320 小時可抵校外實習選修 2 學分，未滿 160 小時不可抵免；實習期滿成績及格者方給予承認課程並抵免學分。學期校外實習之實施為開設 9 學分，至少為期 4.5 個月之校外實習課程，修讀實習課程期間，除依各校訂定定期返校之座談會或研習活動等外，學生應全職於實習機構實習。若於四年級選修學期校外實習課程,則可抵免暑期校外實習及專業選修 7 學分。The implementation of summer off-campus internship is 160 hours and 2 credits in 4 weeks during the summer vacation. If it exceeds 160 hours, the 161-320 hours can be used as 2 credits of off-campus internship. Less than 160 hours cannot be deducted. Only those who have passed the internship period can Grant recognized courses and credit. The implementation of off-campus internship during the semester is an off-campus internship course with 9 credits and a minimum duration of 4.5 months. During the internship course, in addition to regular symposiums or study activities determined by each school, students should practice full-time in the internship institution. . If you take a semester off-campus internship course as an elective in your fourth year, you can deduct 7 credits of summer off-campus internship and professional elective courses.